

# Groundwater storage: modeled outcome levels across 28 management strategies and five hydroclimates

*A summary brief reporting modeled outcome levels for one key outcome under every management strategy in the library, read across the five hydroclimates. Each difference from the baseline is split into the hydroclimate effect and the management effect. Run identifiers for every scenario are listed in the references note.*

**Summary.** This brief reports modeled groundwater storage outcome levels under each of the 28 management strategies and the five hydroclimates in the COEQWAL library. Figure 1 places every result beside the baseline, current operations under the historical hydroclimate, and splits each difference into the hydroclimate effect and the management effect. The table beneath the figure gives the same values as numbers, and the companion data file holds all of them. Cells outlined in the table are those where the two effects run opposite ways; the largest such offset is *Groundwater pumping limits via reduced crop acreage in the Central Valley* under high climate stress, where the hydroclimate effect is +0.3 and the management effect is -0.4.

## What this is

This brief reports modeled groundwater storage outcome levels, one of the nine key outcomes the COEQWAL outcome-level framework defines. It covers all 28 strategies across all five modeled hydroclimates in the COEQWAL library. Every key outcome has a brief of this form, built around that key outcome's results. The strategy summary briefs show one strategy across all nine key outcomes. This brief turns that around: one key outcome across all 28 strategies. A scenario is one strategy under one hydroclimate; the five hydroclimates are a sample of stress levels, not a forecast, and the hydroclimates reference brief describes how they were built.

## How to read the figure

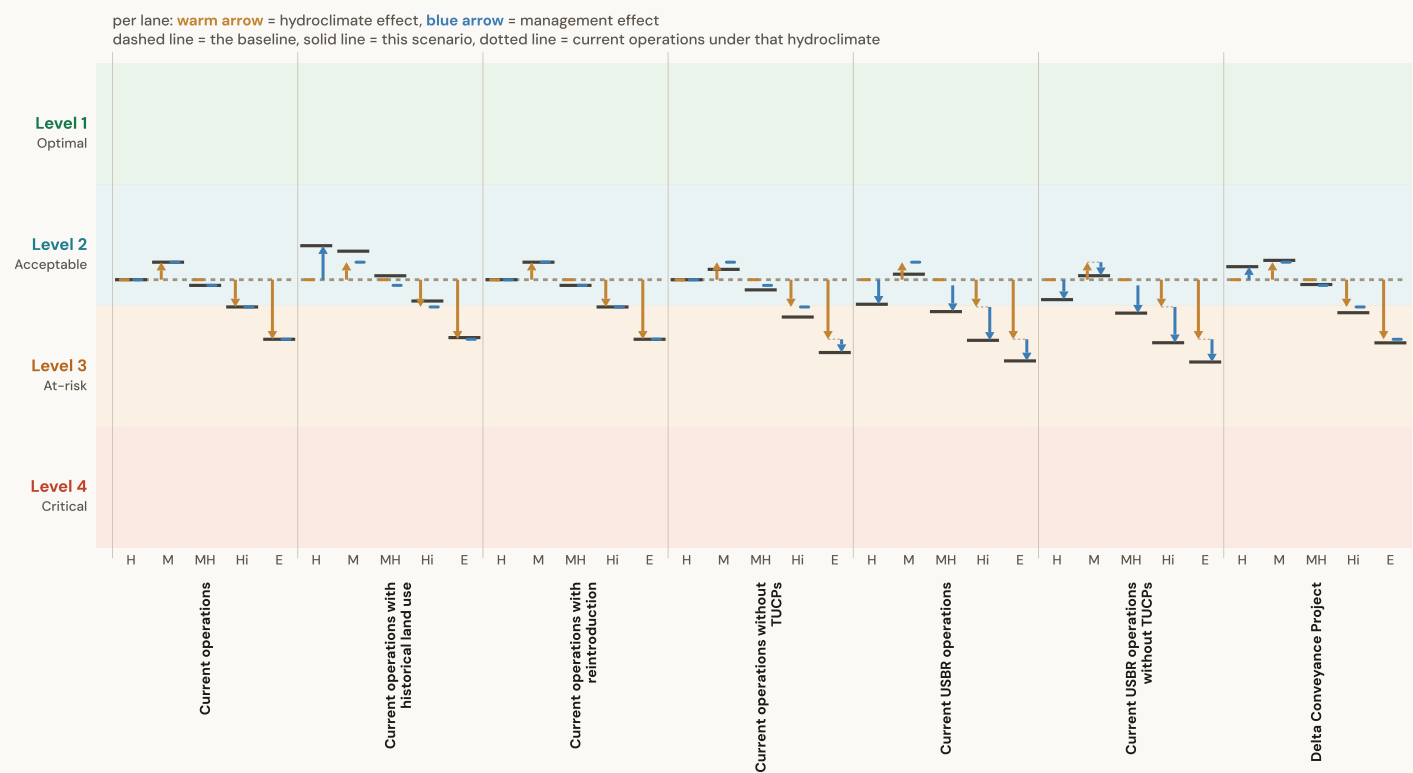
Each mark in Figure 1 is a groundwater storage outcome level under one strategy and one hydroclimate. The baseline value for this key outcome is 2.7, the same reference in every column. Each difference from the baseline is drawn as two arrows. The warm arrow is the hydroclimate effect: how much current operations under that hydroclimate differ from the baseline. The blue arrow is the management effect: how much the strategy differs from current operations under the same hydroclimate. The two add to the difference from the baseline; a flat tick is a change smaller than 0.1.

Values sit on a continuous axis from 1 to 4.99 with Level 1 (Optimal) at the top, and the shaded bands are the four levels. A positive effect moves a value toward Level 4 (Critical); a negative effect moves it toward Level 1 (Optimal). This key outcome's value is the unweighted mean of its 42 locations, so a mark can sit close to the baseline even when the locations behind it differ in both directions; the table 'Locations behind the means' shows how many differ, and in which direction.

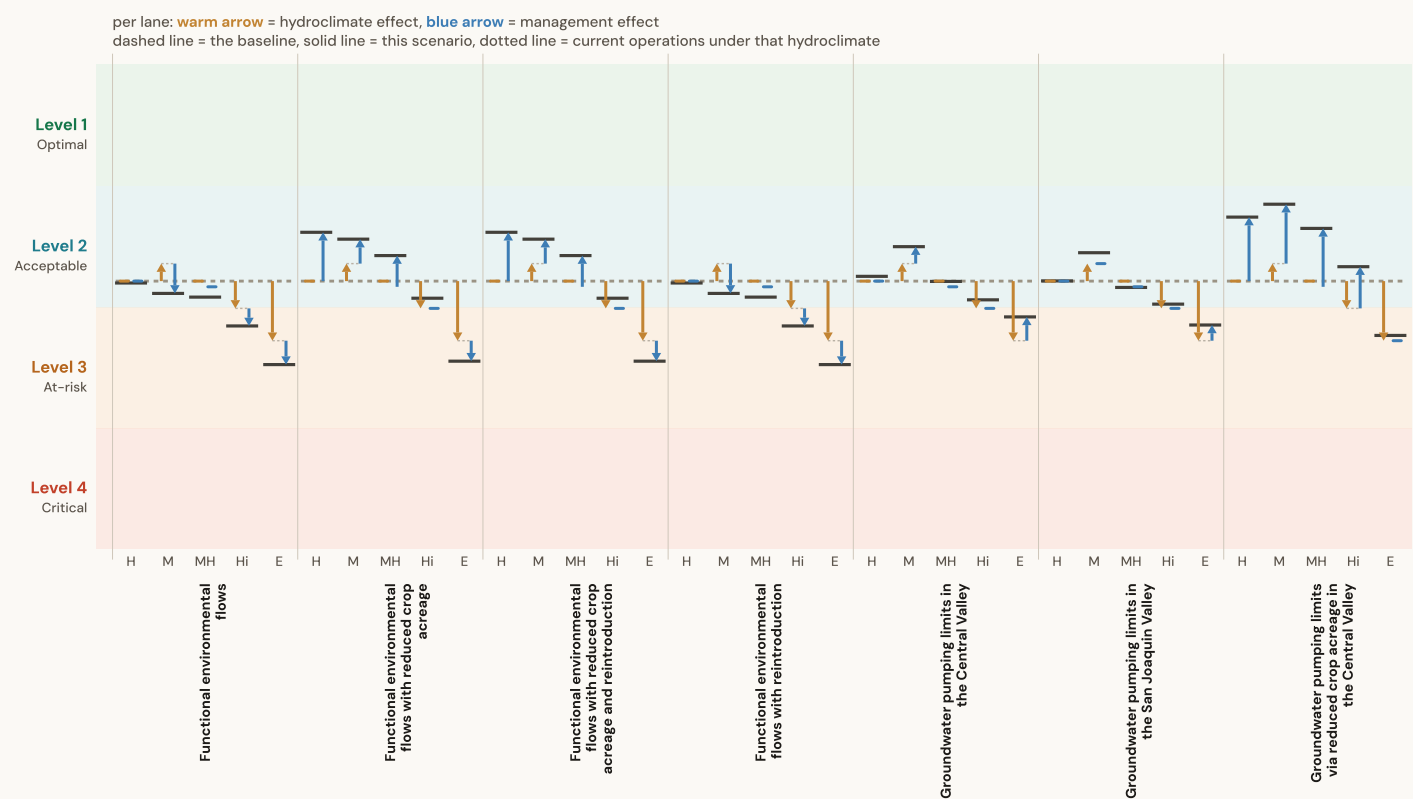
How the split works is explained in Separating the effects of strategy and hydroclimate in COEQWAL results; the scale, thresholds, and rounding rules are in How to read the values below.

# All 28 strategies, all five hydroclimates

Panel 1 of 4 — strategies 1 to 7 of 28

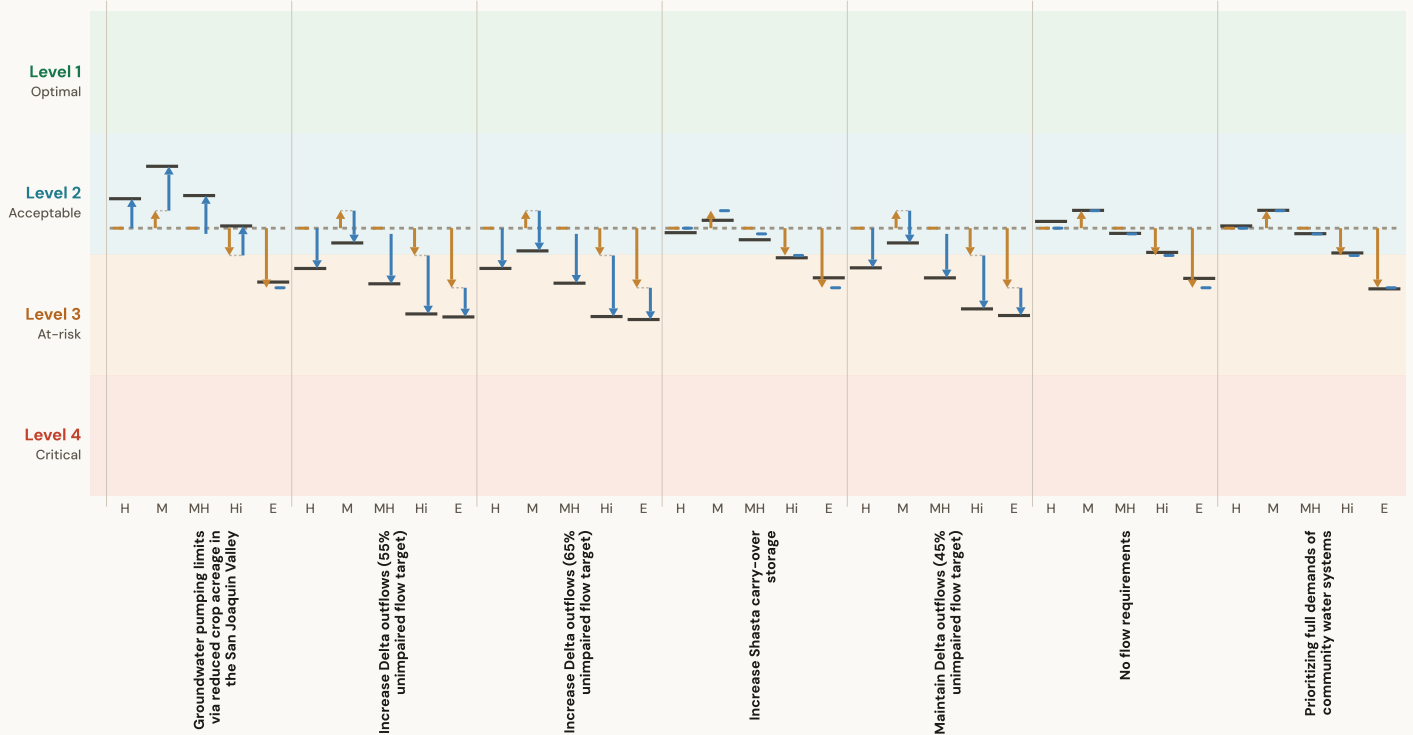


Panel 2 of 4 — strategies 8 to 14 of 28



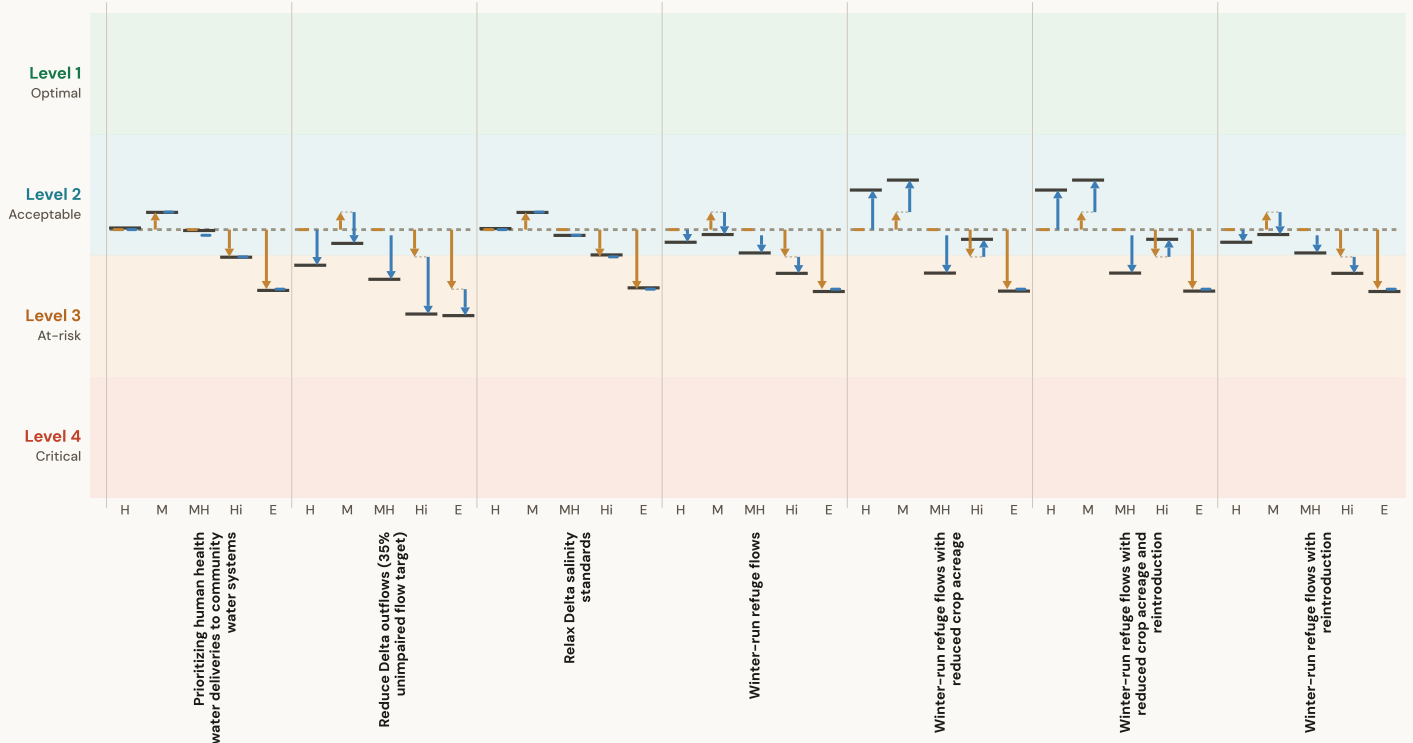
### Panel 3 of 4 — strategies 15 to 21 of 28

per lane: **warm arrow** = hydroclimate effect, **blue arrow** = management effect  
dashed line = the baseline, solid line = this scenario, dotted line = current operations under that hydroclimate



### Panel 4 of 4 — strategies 22 to 28 of 28

per lane: **warm arrow** = hydroclimate effect, **blue arrow** = management effect  
dashed line = the baseline, solid line = this scenario, dotted line = current operations under that hydroclimate



The 28 strategies are shown in four stacked panels that share one level axis. The strategies are split across four panels only to fit the page; read them as one figure.

Figure 1. Modeled level values for groundwater storage and the two effects that produce their difference from the baseline, 28 management strategies × five hydroclimates. Lanes within each strategy run left to right: Historical (H), Moderate (M), Moderate-

high (MH), High (Hi), Extreme (E) climate stress. Vertical axis: continuous level value, Level 1 at top, level bands as shaded zones. Dashed level: the baseline. Solid line: this scenario. A short dotted line where the warm arrow ends and the blue arrow begins marks current operations under that hydroclimate. A short flat tick in place of an arrow is a change smaller than 0.1 on the level scale. Every value is in the table that follows, and the scale is set out in how to read the values.

## The same 140 results as numbers

Each cell reads top to bottom: the hydroclimate effect, the management effect, and the resulting value with its band. The two effects add to the difference between the scenario and the baseline. Outlined cells are those where the two run opposite ways. Readers who want to cite a specific column can link to it directly, and the same numbers appear in the companion data file.

| Strategy                                                                    |                                                           | The baseline | Historical            | Moderate               | Moderate-high         | High                   | Extreme                | Management effect across the five     |
|-----------------------------------------------------------------------------|-----------------------------------------------------------|--------------|-----------------------|------------------------|-----------------------|------------------------|------------------------|---------------------------------------|
| Current operations                                                          | Hydroclimate effect<br>Management effect<br>Value (level) | L2 2.7       | 0.0<br>0.0<br>L2 2.7  | -0.1<br>0.0<br>L2 2.6  | 0.0<br>0.0<br>L2 2.8  | +0.3<br>0.0<br>L3 3.0  | +0.5<br>0.0<br>L3 3.2  | no change at the 0.1 threshold<br>0.0 |
| Current operations with historical land use                                 | Hydroclimate effect<br>Management effect<br>Value (level) | L2 2.7       | 0.0<br>-0.2<br>L2 2.5 | -0.1<br>0.0<br>L2 2.5  | 0.0<br>0.0<br>L2 2.7  | +0.3<br>0.0<br>L2 2.9  | +0.5<br>0.0<br>L3 3.2  | one direction<br>-0.2 to 0.0          |
| Current operations with reintroduction                                      | Hydroclimate effect<br>Management effect<br>Value (level) | L2 2.7       | 0.0<br>0.0<br>L2 2.7  | -0.1<br>0.0<br>L2 2.6  | 0.0<br>0.0<br>L2 2.8  | +0.3<br>0.0<br>L3 3.0  | +0.5<br>0.0<br>L3 3.2  | no change at the 0.1 threshold<br>0.0 |
| Current operations without TUCPs                                            | Hydroclimate effect<br>Management effect<br>Value (level) | L2 2.7       | 0.0<br>0.0<br>L2 2.7  | -0.1<br>0.0<br>L2 2.6  | 0.0<br>0.0<br>L2 2.8  | +0.3<br>0.0<br>L3 3.0  | +0.5<br>+0.1<br>L3 3.3 | one direction<br>0.0 to +0.1          |
| Current USBR operations                                                     | Hydroclimate effect<br>Management effect<br>Value (level) | L2 2.7       | 0.0<br>+0.2<br>L2 2.9 | -0.1<br>0.0<br>L2 2.7  | 0.0<br>+0.2<br>L3 3.0 | +0.3<br>+0.2<br>L3 3.2 | +0.5<br>+0.2<br>L3 3.4 | one direction<br>0.0 to +0.2          |
| Current USBR operations without TUCPs                                       | Hydroclimate effect<br>Management effect<br>Value (level) | L2 2.7       | 0.0<br>+0.2<br>L2 2.9 | -0.1<br>+0.1<br>L2 2.7 | 0.0<br>+0.2<br>L3 3.0 | +0.3<br>+0.3<br>L3 3.3 | +0.5<br>+0.2<br>L3 3.4 | one direction<br>+0.1 to +0.3         |
| Delta Conveyance Project                                                    | Hydroclimate effect<br>Management effect<br>Value (level) | L2 2.7       | 0.0<br>-0.1<br>L2 2.6 | -0.1<br>0.0<br>L2 2.6  | 0.0<br>0.0<br>L2 2.8  | +0.3<br>0.0<br>L3 3.0  | +0.5<br>0.0<br>L3 3.3  | one direction<br>-0.1 to 0.0          |
| Functional environmental flows                                              | Hydroclimate effect<br>Management effect<br>Value (level) | L2 2.7       | 0.0<br>0.0<br>L2 2.8  | -0.1<br>+0.2<br>L2 2.8 | 0.0<br>0.0<br>L2 2.9  | +0.3<br>+0.1<br>L3 3.1 | +0.5<br>+0.2<br>L3 3.4 | one direction<br>0.0 to +0.2          |
| Functional environmental flows with reduced crop acreage                    | Hydroclimate effect<br>Management effect<br>Value (level) | L2 2.7       | 0.0<br>-0.4<br>L2 2.3 | -0.1<br>-0.2<br>L2 2.4 | 0.0<br>-0.3<br>L2 2.5 | +0.3<br>0.0<br>L2 2.9  | +0.5<br>+0.2<br>L3 3.4 | crosses zero<br>-0.4 to +0.2          |
| Functional environmental flows with reduced crop acreage and reintroduction | Hydroclimate effect<br>Management effect<br>Value (level) | L2 2.7       | 0.0<br>-0.4<br>L2 2.3 | -0.1<br>-0.2<br>L2 2.4 | 0.0<br>-0.3<br>L2 2.5 | +0.3<br>0.0<br>L2 2.9  | +0.5<br>+0.2<br>L3 3.4 | crosses zero<br>-0.4 to +0.2          |

| Strategy                                                                             |                                                           | The baseline        | Historical                         | Moderate                            | Moderate-high                      | High                                | Extreme                             | Management effect across the five     |
|--------------------------------------------------------------------------------------|-----------------------------------------------------------|---------------------|------------------------------------|-------------------------------------|------------------------------------|-------------------------------------|-------------------------------------|---------------------------------------|
| <b>Functional environmental flows with reintroduction</b>                            | Hydroclimate effect<br>Management effect<br>Value (level) | <span>L2</span> 2.7 | <span>L2</span> 2.8<br>0.0<br>0.0  | <span>L2</span> 2.8<br>-0.1<br>+0.2 | <span>L2</span> 2.9<br>0.0<br>0.0  | <span>L3</span> 3.1<br>+0.3<br>+0.1 | <span>L3</span> 3.4<br>+0.5<br>+0.2 | one direction<br>0.0 to +0.2          |
| <b>Groundwater pumping limits in the Central Valley</b>                              | Hydroclimate effect<br>Management effect<br>Value (level) | <span>L2</span> 2.7 | <span>L2</span> 2.7<br>0.0<br>0.0  | <span>L2</span> 2.5<br>-0.1<br>-0.1 | <span>L2</span> 2.7<br>0.0<br>0.0  | <span>L2</span> 2.9<br>+0.3<br>0.0  | <span>L3</span> 3.0<br>+0.5<br>-0.2 | one direction<br>-0.2 to 0.0          |
| <b>Groundwater pumping limits in the San Joaquin Valley</b>                          | Hydroclimate effect<br>Management effect<br>Value (level) | <span>L2</span> 2.7 | <span>L2</span> 2.7<br>0.0<br>0.0  | <span>L2</span> 2.5<br>-0.1<br>0.0  | <span>L2</span> 2.8<br>0.0<br>0.0  | <span>L2</span> 2.9<br>+0.3<br>0.0  | <span>L3</span> 3.1<br>+0.5<br>-0.1 | one direction<br>-0.1 to 0.0          |
| <b>Groundwater pumping limits via reduced crop acreage in the Central Valley</b>     | Hydroclimate effect<br>Management effect<br>Value (level) | <span>L2</span> 2.7 | <span>L2</span> 2.2<br>0.0<br>-0.5 | <span>L2</span> 2.1<br>-0.1<br>-0.5 | <span>L2</span> 2.3<br>0.0<br>-0.5 | <span>L2</span> 2.6<br>+0.3<br>-0.4 | <span>L3</span> 3.2<br>+0.5<br>0.0  | one direction<br>-0.5 to 0.0          |
| <b>Groundwater pumping limits via reduced crop acreage in the San Joaquin Valley</b> | Hydroclimate effect<br>Management effect<br>Value (level) | <span>L2</span> 2.7 | <span>L2</span> 2.5<br>0.0<br>-0.2 | <span>L2</span> 2.2<br>-0.1<br>-0.4 | <span>L2</span> 2.5<br>0.0<br>-0.3 | <span>L2</span> 2.7<br>+0.3<br>-0.3 | <span>L3</span> 3.2<br>+0.5<br>0.0  | one direction<br>-0.4 to 0.0          |
| <b>Increase Delta outflows (55% unimpaired flow target)</b>                          | Hydroclimate effect<br>Management effect<br>Value (level) | <span>L2</span> 2.7 | <span>L3</span> 3.1<br>0.0<br>+0.4 | <span>L2</span> 2.9<br>-0.1<br>+0.3 | <span>L3</span> 3.2<br>0.0<br>+0.4 | <span>L3</span> 3.4<br>+0.3<br>+0.4 | <span>L3</span> 3.5<br>+0.5<br>+0.3 | one direction<br>+0.3 to +0.4         |
| <b>Increase Delta outflows (65% unimpaired flow target)</b>                          | Hydroclimate effect<br>Management effect<br>Value (level) | <span>L2</span> 2.7 | <span>L3</span> 3.1<br>0.0<br>+0.4 | <span>L2</span> 2.9<br>-0.1<br>+0.3 | <span>L3</span> 3.2<br>0.0<br>+0.4 | <span>L3</span> 3.5<br>+0.3<br>+0.5 | <span>L3</span> 3.5<br>+0.5<br>+0.3 | one direction<br>+0.3 to +0.5         |
| <b>Increase Shasta carry-over storage</b>                                            | Hydroclimate effect<br>Management effect<br>Value (level) | <span>L2</span> 2.7 | <span>L2</span> 2.8<br>0.0<br>0.0  | <span>L2</span> 2.7<br>-0.1<br>0.0  | <span>L2</span> 2.8<br>0.0<br>0.0  | <span>L3</span> 3.0<br>+0.3<br>0.0  | <span>L3</span> 3.1<br>+0.5<br>0.0  | no change at the 0.1 threshold<br>0.0 |

| Strategy                                                              |                                                           | The baseline      | Historical                       | Moderate                          | Moderate-high                    | High                              | Extreme                           | Management effect across the five     |
|-----------------------------------------------------------------------|-----------------------------------------------------------|-------------------|----------------------------------|-----------------------------------|----------------------------------|-----------------------------------|-----------------------------------|---------------------------------------|
| Maintain Delta outflows (45% unimpaired flow target)                  | Hydroclimate effect<br>Management effect<br>Value (level) | <div>L2</div> 2.7 | <div>L3</div> 0.0<br>+0.4<br>3.1 | <div>L2</div> -0.1<br>+0.3<br>2.9 | <div>L3</div> 0.0<br>+0.3<br>3.1 | <div>L3</div> +0.3<br>+0.4<br>3.4 | <div>L3</div> +0.5<br>+0.3<br>3.5 | one direction<br>+0.3 to +0.4         |
| No flow requirements                                                  | Hydroclimate effect<br>Management effect<br>Value (level) | <div>L2</div> 2.7 | <div>L2</div> 0.0<br>0.0<br>2.7  | <div>L2</div> -0.1<br>0.0<br>2.6  | <div>L2</div> 0.0<br>0.0<br>2.8  | <div>L2</div> +0.3<br>0.0<br>2.9  | <div>L3</div> +0.5<br>0.0<br>3.1  | no change at the 0.1 threshold<br>0.0 |
| Prioritizing full demands of community water systems                  | Hydroclimate effect<br>Management effect<br>Value (level) | <div>L2</div> 2.7 | <div>L2</div> 0.0<br>0.0<br>2.7  | <div>L2</div> -0.1<br>0.0<br>2.6  | <div>L2</div> 0.0<br>0.0<br>2.8  | <div>L2</div> +0.3<br>0.0<br>2.9  | <div>L3</div> +0.5<br>0.0<br>3.2  | no change at the 0.1 threshold<br>0.0 |
| Prioritizing human health water deliveries to community water systems | Hydroclimate effect<br>Management effect<br>Value (level) | <div>L2</div> 2.7 | <div>L2</div> 0.0<br>0.0<br>2.7  | <div>L2</div> -0.1<br>0.0<br>2.6  | <div>L2</div> 0.0<br>0.0<br>2.7  | <div>L3</div> +0.3<br>0.0<br>3.0  | <div>L3</div> +0.5<br>0.0<br>3.2  | no change at the 0.1 threshold<br>0.0 |
| Reduce Delta outflows (35% unimpaired flow target)                    | Hydroclimate effect<br>Management effect<br>Value (level) | <div>L2</div> 2.7 | <div>L3</div> 0.0<br>+0.3<br>3.0 | <div>L2</div> -0.1<br>+0.2<br>2.8 | <div>L3</div> 0.0<br>+0.3<br>3.1 | <div>L3</div> +0.3<br>+0.4<br>3.4 | <div>L3</div> +0.5<br>+0.2<br>3.4 | one direction<br>+0.2 to +0.4         |
| Relax Delta salinity standards                                        | Hydroclimate effect<br>Management effect<br>Value (level) | <div>L2</div> 2.7 | <div>L2</div> 0.0<br>0.0<br>2.7  | <div>L2</div> -0.1<br>0.0<br>2.6  | <div>L2</div> 0.0<br>0.0<br>2.8  | <div>L2</div> +0.3<br>0.0<br>2.9  | <div>L3</div> +0.5<br>0.0<br>3.2  | no change at the 0.1 threshold<br>0.0 |
| Winter-run refuge flows                                               | Hydroclimate effect<br>Management effect<br>Value (level) | <div>L2</div> 2.7 | <div>L2</div> 0.0<br>+0.1<br>2.8 | <div>L2</div> -0.1<br>+0.2<br>2.8 | <div>L2</div> 0.0<br>+0.1<br>2.9 | <div>L3</div> +0.3<br>+0.1<br>3.1 | <div>L3</div> +0.5<br>0.0<br>3.2  | one direction<br>0.0 to +0.2          |
| Winter-run refuge flows with reduced crop acreage                     | Hydroclimate effect<br>Management effect<br>Value (level) | <div>L2</div> 2.7 | <div>L2</div> 0.0<br>-0.3<br>2.4 | <div>L2</div> -0.1<br>-0.3<br>2.3 | <div>L3</div> 0.0<br>+0.3<br>3.1 | <div>L2</div> +0.3<br>-0.2<br>2.8 | <div>L3</div> +0.5<br>0.0<br>3.2  | crosses zero<br>-0.3 to +0.3          |
| Winter-run refuge flows with reduced crop acreage and reintroduction  | Hydroclimate effect<br>Management effect<br>Value (level) | <div>L2</div> 2.7 | <div>L2</div> 0.0<br>-0.3<br>2.4 | <div>L2</div> -0.1<br>-0.3<br>2.3 | <div>L3</div> 0.0<br>+0.3<br>3.1 | <div>L2</div> +0.3<br>-0.2<br>2.8 | <div>L3</div> +0.5<br>0.0<br>3.2  | crosses zero<br>-0.3 to +0.3          |
| Winter-run refuge flows                                               | Hydroclimate effect<br>Management effect<br>Value (level) | <div>L2</div> 2.7 | <div>L2</div> 0.0<br>+0.1<br>2.8 | <div>L2</div> -0.1<br>+0.2<br>2.8 | <div>L2</div> 0.0<br>+0.1<br>2.9 | <div>L3</div> +0.3<br>+0.1<br>3.1 | <div>L3</div> +0.5<br>0.0<br>3.2  | one direction<br>0.0 to +0.2          |

| Strategy            |  | The baseline | Historical | Moderate | Moderate-high | High | Extreme | Management effect across the five |
|---------------------|--|--------------|------------|----------|---------------|------|---------|-----------------------------------|
| with reintroduction |  |              |            |          |               |      |         |                                   |

## Locations behind the means

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A key outcome’s value is the mean of its locations, so one mean can stand for locations that moved in different directions. The table counts, for each strategy and hydroclimate, how many of this key outcome’s 42 locations differ from current operations by at least 0.1 on the level scale, and in which direction. To see the data for individual locations, view the [Locations of Interest summary briefs](#).

| Strategy                                                                    |                | Historical | Moderate | Moderate-high | High | Extreme |
|-----------------------------------------------------------------------------|----------------|------------|----------|---------------|------|---------|
| Current operations                                                          | Toward Level 1 | 0          | 0        | 0             | 0    | 0       |
|                                                                             | No difference  | 42         | 42       | 42            | 42   | 42      |
|                                                                             | Toward Level 4 | 0          | 0        | 0             | 0    | 0       |
| Current operations with historical land use                                 | Toward Level 1 | 21         | 12       | 12            | 13   | 11      |
|                                                                             | No difference  | 19         | 19       | 23            | 23   | 21      |
|                                                                             | Toward Level 4 | 2          | 11       | 7             | 6    | 10      |
| Current operations with reintroduction                                      | Toward Level 1 | 0          | 0        | 0             | 0    | 0       |
|                                                                             | No difference  | 42         | 42       | 42            | 42   | 42      |
|                                                                             | Toward Level 4 | 0          | 0        | 0             | 0    | 0       |
| Current operations without TUCPs                                            | Toward Level 1 | 0          | 0        | 0             | 1    | 0       |
|                                                                             | No difference  | 40         | 31       | 36            | 35   | 30      |
|                                                                             | Toward Level 4 | 2          | 11       | 6             | 6    | 12      |
| Current USBR operations                                                     | Toward Level 1 | 16         | 19       | 16            | 18   | 20      |
|                                                                             | No difference  | 4          | 4        | 5             | 3    | 3       |
|                                                                             | Toward Level 4 | 22         | 19       | 21            | 21   | 19      |
| Current USBR operations without TUCPs                                       | Toward Level 1 | 16         | 19       | 16            | 17   | 19      |
|                                                                             | No difference  | 4          | 3        | 5             | 3    | 4       |
|                                                                             | Toward Level 4 | 22         | 20       | 21            | 22   | 19      |
| Delta Conveyance Project                                                    | Toward Level 1 | 15         | 4        | 4             | 3    | 9       |
|                                                                             | No difference  | 24         | 30       | 32            | 31   | 19      |
|                                                                             | Toward Level 4 | 3          | 8        | 6             | 8    | 14      |
| Functional environmental flows                                              | Toward Level 1 | 6          | 2        | 2             | 2    | 1       |
|                                                                             | No difference  | 25         | 17       | 23            | 17   | 19      |
|                                                                             | Toward Level 4 | 11         | 23       | 17            | 23   | 22      |
| Functional environmental flows with reduced crop acreage                    | Toward Level 1 | 23         | 19       | 18            | 16   | 3       |
|                                                                             | No difference  | 12         | 11       | 15            | 14   | 20      |
|                                                                             | Toward Level 4 | 7          | 12       | 9             | 12   | 19      |
| Functional environmental flows with reduced crop acreage and reintroduction | Toward Level 1 | 23         | 19       | 18            | 16   | 3       |
|                                                                             | No difference  | 12         | 11       | 15            | 14   | 20      |
|                                                                             | Toward Level 4 | 7          | 12       | 9             | 12   | 19      |
| Functional environmental flows with reintroduction                          | Toward Level 1 | 6          | 2        | 2             | 2    | 1       |
|                                                                             | No difference  | 25         | 17       | 23            | 17   | 19      |
|                                                                             | Toward Level 4 | 11         | 23       | 17            | 23   | 22      |
| Groundwater pumping limits in the Central Valley                            | Toward Level 1 | 14         | 12       | 12            | 14   | 21      |
|                                                                             | No difference  | 20         | 24       | 21            | 22   | 18      |
|                                                                             | Toward Level 4 | 8          | 6        | 9             | 6    | 3       |
| Groundwater pumping limits in the San Joaquin Valley                        | Toward Level 1 | 7          | 7        | 6             | 7    | 14      |
|                                                                             | No difference  | 29         | 25       | 25            | 28   | 24      |
|                                                                             | Toward Level 4 | 6          | 10       | 11            | 7    | 4       |

| Strategy                                                                      |                | Historical | Moderate | Moderate-high | High | Extreme |
|-------------------------------------------------------------------------------|----------------|------------|----------|---------------|------|---------|
| Groundwater pumping limits via reduced crop acreage in the Central Valley     | Toward Level 1 | 23         | 22       | 23            | 28   | 12      |
|                                                                               | No difference  | 11         | 15       | 10            | 10   | 27      |
|                                                                               | Toward Level 4 | 8          | 5        | 9             | 4    | 3       |
| Groundwater pumping limits via reduced crop acreage in the San Joaquin Valley | Toward Level 1 | 11         | 15       | 14            | 20   | 12      |
|                                                                               | No difference  | 31         | 27       | 28            | 22   | 27      |
|                                                                               | Toward Level 4 | 0          | 0        | 0             | 0    | 3       |
| Increase Delta outflows (55% unimpaired flow target)                          | Toward Level 1 | 15         | 18       | 16            | 16   | 18      |
|                                                                               | No difference  | 4          | 3        | 4             | 4    | 5       |
|                                                                               | Toward Level 4 | 23         | 21       | 22            | 22   | 19      |
| Increase Delta outflows (65% unimpaired flow target)                          | Toward Level 1 | 15         | 17       | 16            | 16   | 18      |
|                                                                               | No difference  | 4          | 3        | 4             | 4    | 4       |
|                                                                               | Toward Level 4 | 23         | 22       | 22            | 22   | 20      |
| Increase Shasta carry-over storage                                            | Toward Level 1 | 2          | 0        | 0             | 6    | 15      |
|                                                                               | No difference  | 34         | 33       | 39            | 30   | 20      |
|                                                                               | Toward Level 4 | 6          | 9        | 3             | 6    | 7       |
| Maintain Delta outflows (45% unimpaired flow target)                          | Toward Level 1 | 15         | 18       | 17            | 16   | 18      |
|                                                                               | No difference  | 4          | 2        | 4             | 4    | 5       |
|                                                                               | Toward Level 4 | 23         | 22       | 21            | 22   | 19      |
| No flow requirements                                                          | Toward Level 1 | 10         | 7        | 8             | 7    | 10      |
|                                                                               | No difference  | 31         | 28       | 28            | 31   | 30      |
|                                                                               | Toward Level 4 | 1          | 7        | 6             | 4    | 2       |
| Prioritizing full demands of community water systems                          | Toward Level 1 | 2          | 0        | 0             | 2    | 0       |
|                                                                               | No difference  | 40         | 42       | 42            | 40   | 40      |
|                                                                               | Toward Level 4 | 0          | 0        | 0             | 0    | 2       |
| Prioritizing human health water deliveries to community water systems         | Toward Level 1 | 2          | 0        | 1             | 0    | 0       |
|                                                                               | No difference  | 40         | 42       | 41            | 42   | 40      |
|                                                                               | Toward Level 4 | 0          | 0        | 0             | 0    | 2       |
| Reduce Delta outflows (35% unimpaired flow target)                            | Toward Level 1 | 14         | 18       | 17            | 16   | 20      |
|                                                                               | No difference  | 4          | 2        | 4             | 4    | 3       |
|                                                                               | Toward Level 4 | 24         | 22       | 21            | 22   | 19      |
| Relax Delta salinity standards                                                | Toward Level 1 | 1          | 0        | 1             | 1    | 1       |
|                                                                               | No difference  | 41         | 42       | 41            | 41   | 41      |
|                                                                               | Toward Level 4 | 0          | 0        | 0             | 0    | 0       |
| Winter-run refuge flows                                                       | Toward Level 1 | 6          | 3        | 3             | 2    | 11      |
|                                                                               | No difference  | 19         | 19       | 14            | 20   | 18      |
|                                                                               | Toward Level 4 | 17         | 20       | 25            | 20   | 13      |

| Strategy                                                             |                | Historical | Moderate | Moderate-high | High | Extreme |
|----------------------------------------------------------------------|----------------|------------|----------|---------------|------|---------|
| Winter-run refuge flows with reduced crop acreage                    | Toward Level 1 | 20         | 17       | 0             | 15   | 11      |
|                                                                      | No difference  | 10         | 10       | 12            | 13   | 18      |
|                                                                      | Toward Level 4 | 12         | 15       | 30            | 14   | 13      |
| Winter-run refuge flows with reduced crop acreage and reintroduction | Toward Level 1 | 20         | 17       | 0             | 15   | 11      |
|                                                                      | No difference  | 10         | 10       | 12            | 13   | 18      |
|                                                                      | Toward Level 4 | 12         | 15       | 30            | 14   | 13      |
| Winter-run refuge flows with reintroduction                          | Toward Level 1 | 6          | 3        | 3             | 2    | 11      |
|                                                                      | No difference  | 19         | 19       | 14            | 20   | 18      |
|                                                                      | Toward Level 4 | 17         | 20       | 25            | 20   | 13      |

Per-location detail and the median difference behind each count are in the companion data file at [#/rows/\\*/reach](#).

## Related briefs

|                                                                                                                                                           |                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| <b>What happens in another key outcome?</b> A key-outcome brief reports one key outcome of the water system, every strategy, the same five hydroclimates: |                                           |
| Agricultural revenue                                                                                                                                      | Freshwater for in-Delta uses              |
| Community surface water                                                                                                                                   | Groundwater storage — <i>you are here</i> |
| Delta estuary ecology                                                                                                                                     | Reservoir storage                         |
| Environmental flows                                                                                                                                       | Winter-run salmon                         |
| Freshwater for Delta exports                                                                                                                              |                                           |

|                                                                                                                                                                                                                                              |                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| <b>What happens across all nine key outcomes under one strategy?</b> This brief holds the key outcome fixed and varies the strategy. The transpose does the reverse, reporting every key outcome under one strategy, one brief per strategy: |                                                                               |
| Current operations                                                                                                                                                                                                                           | Groundwater pumping limits via reduced crop acreage in the Central Valley     |
| Current operations with historical land use                                                                                                                                                                                                  | Groundwater pumping limits via reduced crop acreage in the San Joaquin Valley |
| Current operations with reintroduction                                                                                                                                                                                                       | Increase Delta outflows (55% unimpaired flow target)                          |
| Current operations without TUCPs                                                                                                                                                                                                             | Increase Delta outflows (65% unimpaired flow target)                          |
| Current USBR operations                                                                                                                                                                                                                      | Increase Shasta carry-over storage                                            |
| Current USBR operations without TUCPs                                                                                                                                                                                                        | Maintain Delta outflows (45% unimpaired flow target)                          |
| Delta Conveyance Project                                                                                                                                                                                                                     | No flow requirements                                                          |
| Functional environmental flows                                                                                                                                                                                                               | Prioritizing full demands of community water systems                          |
| Functional environmental flows with reduced crop acreage                                                                                                                                                                                     | Prioritizing human health water deliveries to community water systems         |
| Functional environmental flows with reduced crop acreage and reintroduction                                                                                                                                                                  | Reduce Delta outflows (35% unimpaired flow target)                            |
| Functional environmental flows with reintroduction                                                                                                                                                                                           | Relax Delta salinity standards                                                |
| Groundwater pumping limits in the Central Valley                                                                                                                                                                                             | Winter-run refuge flows                                                       |
| Groundwater pumping limits in the San Joaquin Valley                                                                                                                                                                                         | Winter-run refuge flows with reduced crop acreage                             |

Winter-run refuge flows with reduced crop acreage and reintroduction

Winter-run refuge flows with reintroduction

Briefs for all locations of interest are available from the COEQWAL Data page.

**How were the five hydroclimates constructed?** The hydroclimates reference brief defines them. It reports no results of its own; every brief above is read against the same five.

## What this doesn't say

This brief reports modeled results from the COEQWAL ensemble. It does not predict what will happen, evaluate whether those results are acceptable, or recommend any course of action. The outcome-level framework provides definitional categories; the meaning of those categories for any individual reader, community, or decision is not specified by COEQWAL.

In particular, for this brief: the five hydroclimates are a sample of stress levels, carry no probabilities, and are not a timeline, so no average over them is reported here. Splitting a difference into two effects is arithmetic on modeled values, not a causal claim about the water system, and the two are separable only because each is defined against its own reference. A step or a spread on the level axis is a modeled magnitude, not a benefit or a cost. A key outcome's value is the unweighted mean of its locations, so locations count equally regardless of the people, acres, or acre-feet each represents, and a location already at either end of the scale cannot move further along it. Locations are modeled nodes, not measured conditions at a place.

A strategy's highest value does not necessarily fall under extreme climate stress.

## References, run identifiers and the build stamp

**Model runs** (current operations): Historical `s0020`; Moderate `s0134`; Moderate-high `s0047`; High `s0056`; Extreme `s0108`. Each row reads its own five runs beside these; they are in the companion data file at `#/rows/*/by_hydroclimate/*/run_ids`. The baseline is current operations under the historical hydroclimate, `s0020`.

**Build stamp.** Snapshot `2026-08-29`; builder version `0.1.0`; built `2026-09-25T20:58:30Z`; snapshot data read `2026-08-29`; verification: pass. The verification status is written by the independent check (`briefbuilder.verifier`), which recomputes every value in the companion data file from the per-location data and reads every arrow on this page, after both files are written. It is not stamped by whatever produced them.

Companion data file: `key_outcome_summary_groundwater-storage_schema1.json`, holding every value in this brief plus the per-location counts, the spread measures, the derivation of every number in the prose and the figure's declared attribute contract.

Per-hydroclimate slices: Historical `#/rows/*/by_hydroclimate/BL`; Moderate `#/rows/*/by_hydroclimate/HC4`; Moderate-high `#/rows/*/by_hydroclimate/HC1`; High `#/rows/*/by_hydroclimate/HC2`; Extreme `#/rows/*/by_hydroclimate/HC3`. Built from the locked 2026-08-29 API snapshot (api.coeqwal.org v2.1.1 (locked dataset snapshot)). Decomposition per

`Decomposition_Math_Quickref` (coeqwal-viz): the hydroclimate effect is current operations under that hydroclimate minus current operations under the historical hydroclimate; the management effect is this strategy minus current operations under that same hydroclimate. Axis convention [1, 4.99] per `Ordinal_vs_continuous_tiers.md`; the visible band is `floor(value)`. A difference of 0.1 on the level scale is required before a difference is reported as movement, before a location is counted as

differing from its comparison, and before an effect prints as anything other than 0.0. Color semantics: warm = hydroclimate, blue = management. Strategy identity is its descriptive name; run IDs appear only in this note and the data file.

## How to read the values

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The level scale runs from 1 to 4.99. Level 1 is the favorable end of the scale: the outcome-level framework names Level 1 Optimal and Level 4 Critical, so a smaller value sits nearer Optimal and a larger value nearer Critical. The API serves this number in a field named `weighted_score`; that name describes how the number is computed, and does not mean that a larger number is a more favorable result.

Two numbers describe each result. The integer `tier_level` is the level label. `tier_continuous` is the position within the band of that label, and always falls inside it: `tier_continuous` is at least `tier_level` and less than `tier_level` plus one. Labels, counts and prose use `tier_level`; figure positions and statistics use `tier_continuous`.

Level 1 is Optimal, Level 2 is Acceptable, Level 3 is At-risk, and Level 4 is Critical. These are definitional category names supplied by the outcome-level framework; what a category means for any reader, community, or decision is not specified by COEQWAL.

A key outcome value is the unweighted mean over the locations in that key outcome. Every location counts equally, regardless of the people, acres, or acre-feet it represents.

Any difference between two values under different hydroclimates contains both a hydroclimate effect and a management effect; the two are separable only through the decomposition, because each is defined against its own reference. A strategy is compared with current operations under the same hydroclimate. Comparing one strategy with itself across hydroclimates is a different quantity: the hydroclimate effect plus the change in the management effect. The hydroclimate effect is always measured on current operations, not on the strategy.

One threshold governs every claim and every count here. A difference smaller than 0.1 on the level scale is not reported as movement, and a location is not counted as differing from its comparison. An outcome level is a summary built so that results in nine key outcomes with no shared unit can be compared, and 0.1 on the level scale is the finest reading it carries; a smaller difference is left unstated rather than described.

The five hydroclimates were chosen to span a range of climate stress levels. They are not a sequence, a trend, or a forecast, they carry no probabilities, and no mean is taken over them.

A run ID identifies one modeling run of one strategy under one hydroclimate. It is an internal identifier, used for citation and for tracing a value back to the data platform, not a name for the strategy; the public identity of a strategy is its descriptive name.

Level values are reported to one decimal place throughout this brief — in the prose, in the figure and in the table. The level scale does not carry meaning finer than that. Each effect shown is the difference between the floored values, except that an effect smaller than 0.1 on the level scale, which is the threshold this brief reports movement at, is shown as 0.0, so a row containing one does not add up. Values are floored rather than rounded, so a printed value stays inside the band its own label names; full-precision values are in the machine-readable sidecar this brief is built from.